Table 2: Read-only audit of OOS artifacts. The retained catalog was selected using OOS outcomes.

Artifact count	Value
Checkpoint expressions	2,016
Negative-OOS checkpoint rows	368
Retained catalog expressions	1,837
Non-positive-OOS catalog rows	0
Checkpoint-only expressions	368
Checkpoint-only negative OOS	368
Catalog rows passing stated IS rule	1,037

preserved artifacts and runner shows that this interpretation is not valid: the runner uses a looser preliminary screen and explicitly drops records with OOS Sharpe ≤ 0 before writing the retained file. Moreover, only 1,037 rows in that file satisfy the manuscript's stated IS rule (Sharpe ≥ 2, Fitness ≥ 0, Turnover ≤ 500). The checkpoint contains 2,016 unique expressions, including 368 with negative OOS Sharpe. All 368 expressions found only in the checkpoint are negative OOS; all 189 expressions found only in the retained catalog are positive OOS; and the 1,648 common expressions have no sign disagreement. The catalog's zero sign-reversal rate and its decay summaries are therefore properties of outcome-based retention, not evidence that the mined factors generalize.

A clean OOS evaluation remains pending. It must recover and hash the complete pre-OOS pool, pre-specify an IS-only screen, perform any deduplication using IS data, freeze the expression list before OOS access, checkpoint every response, and preserve every numerical OOS outcome and error regardless of sign. Numerical OOS and generalization claims should not be restored until that run exists.

6 Full-Cohort Public Reanalysis

We separately evaluate a frozen list of 1,047 unique expressions selected from 3,114 preserved input rows using the stated IS screen. This does not establish a census of all generated candidates or a historically blind holdout. The broader preserved pool is not a matched replay of the three historical keep totals. All IDs are retained regardless of subsequent public-evaluator outcome; no expression is chosen by its new score.

The new snapshot contains the same ordered 40-stock convenience universe over 727 trading dates in 2023–2025. We acquire both unadjusted and provider-adjusted daily views through BaoStock 0.9.3 and verify all 80 files, their byte hashes,

Table 3: Full frozen cohort under the corrected public evaluator. Non-scorable records remain in the denominator; no missing result is assigned zero alpha.

Outcome	Records
Scorable signed rank-IC	923
Insufficient valid dates	99
Unsupported operator or variable	25
All frozen records	**1,047**
Positive mean rank-IC (of 923)	671
Negative mean rank-IC (of 923)	252
Median mean rank-IC (of 923)	0.0148

date/stock/adjustment identities, and shared quantity and trading-status fields. Features use nominal prices, shares, and nominal traded amount; the adjusted-price view is reserved for labels. Two non-trading stock-day cells are masked. We neither forward-fill missing observations nor substitute a smaller acquired stock prefix.

For each 2024–2025 date t, the label is the provider-adjusted close change over five trading-calendar rows,

$$y_{i,t} = \frac{P^{\text{adj}}_{i,t+5}}{P^{\text{adj}}_{i,t}} - 1.$$

Both endpoint prices must be finite and positive and both endpoints tradable. This is not an investor total-return or executable-P&L claim; the last five captured dates have no label. Daily rank-IC is the Pearson correlation of average-tie cross-sectional ranks. We require at least five eligible stocks, three distinct signal values, and two distinct labels. The three-signal-value threshold is a chosen diagnostic convention, not a claim that binary signals cannot have a mathematically defined correlation. A factor is scorable only with at least 20 valid dates. Its mean/SD ratio, if reported, is unannualized and is not a trading Sharpe.

The frozen public implementation initially produces 57 numerical errors: its conditional operator attempts to interpret a panel as one Boolean when the true branch is scalar. We preserve this v1 run and add a versioned scalar-broadcasting correction with the existing elementwise conditional semantics. All 1,047 records are then re-evaluated, rather than rerunning only promising failures. The other 990 diagnostic records remain unchanged; 56 corrected records become scorable and one remains insufficiently observed.

Table 3 is descriptive. Of the 923 scorable expressions, 835 have all 480 label-available target